

Nicolas Gioanni

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EDUCATION

University of Washington M.S. in Computer Science & Software Engineering	Jun 2028
University of Washington B.S. in Computer Science & Software Engineering (Concentration: Information Assurance & Cybersecurity)	Jun 2025 GPA: 3.75/4.0
• Dean's List (Fall 2022 – Spring 2025) • Relevant Coursework: Data Structures & Algorithms, Advanced Algorithms, Operating Systems, Secure Systems, Database Systems, Software Engineering, Network Design & Programming, Technical Writing	

EXPERIENCE

UW Bothell School of STEM <i>Research Assistant (Software Engineering)</i>	Aug 2024 – Present Bothell, WA
• Architected and deployed a full-stack computer-vision web app for the University of Utah Miller Lab using Python, Django, REST APIs, and a JavaScript frontend with Fetch API, reducing scientists' manual microscopy analysis by 97% • Built and optimized a deterministic batch ML pipeline using TensorFlow/Keras, OpenCV, scikit-image, and NumPy to automate Mask R-CNN segmentation and extract 30+ cell-level metrics, cutting processing time by 73% and eliminating manual errors • Re-architected the data layer by migrating SQLite to PostgreSQL, removing redundant previews, and converting preprocessing caches from TIFF to PNG, reducing file storage by 56.6% and increasing 1 GB quota from 37 to 86 files • Led a 6-person development team in an Agile workflow, maintaining CI/CD pipelines and deploying the platform on a UW-IT Linux VM with Nginx, Unicorn, and Let's Encrypt TLS for use by 15+ scientists	
University of Washington <i>Teaching Assistant</i>	Jan 2025 – Jun 2025 Bothell, WA
• Led weekly office hours and quiz sections, provided 1:1 tutoring, and managed grading for 600+ students in Data Structures & Algorithms, Discrete Mathematics, Operating Systems, and Software Engineering using C/C++, JavaScript, OOP, Git, and Docker • Evaluated coding projects for 400+ students through code review, achieving an average student rating of 4.91/5	
University of Washington <i>Undergraduate Researcher (Adversarial Machine Learning)</i>	Sep 2024 – Dec 2024 Bothell, WA
• Implemented adversarial attack and defense pipelines for image classification models in Python using TensorFlow/Keras • Optimized defense execution with batched Keras evaluation to under 5 seconds while preserving 94% clean-data accuracy • Utilized NumPy and Matplotlib to automate attack and defense testing, tracking success rates, latency, and accuracy	
UW Bothell School of STEM <i>Research Assistant (AI/ML)</i>	Mar 2024 – Aug 2024 Bothell, WA
• Developed an automated system performing target selection in yeast DNA for researchers, reducing manual workflows by 99% • Used Python, Pandas, Biopython, and regex to analyze 18,000+ FASTA files and export validated sequences to Excel	

PROJECTS

NotePal (Note-Taking & Study App)	github.com/nicolasgioanni/NotePal
• Built and deployed a full-stack study app on Vercel with CI/CD using Next.js/TypeScript and OpenAI API, featuring a Notion style editor and AI autocomplete to generate notes, flashcards, and quizzes from multimodal input, cutting study time by 67% • Engineered a modular Flask REST API backend that orchestrates OCR, ASR, and LLM pipelines with Whisper, YOLO, and OpenAI APIs, securing per-user data with Firebase Auth and Firestore, and enables RAG chat backed by Pinecone vector search	
Clair (Automotive File Organizer)	github.com/nicolasgioanni/Clair
• Built a file organization desktop app with Python and Bash to automate folder cleanup, reducing manual sorting time by 80%	

LEADERSHIP

Google Developer Groups on Campus Officer	Sep 2024 – Jun 2025 Bothell, WA
• Mentored 100+ students through workshops on courses, coding practices, and tools including Git, Docker, and VS Code	

MANUSCRIPTS IN PREPARATION

Gioanni, Nicolas. First-author manuscript in preparation for Elsevier's *SoftwareX* on CytoCV with the University of Utah Miller Lab

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Bash, C/C++, HTML/CSS, Assembly
Frameworks: Django, Flask, Node.js, Next.js, React
ML/CV & Data: TensorFlow/Keras, OpenCV, scikit-image, NumPy, pandas, Matplotlib, Pillow
Cloud/Dev Tools: Git, Docker, Linux, SSH, Nginx, Unicorn, Let's Encrypt, PostgreSQL, Google Cloud, Firestore, Firebase, Azure